

Supplemental Methods: Data Availability and Access

1 Overview

All raw data, preprocessed training subsets, trained models, learned embeddings, and mutual-information (MI) estimates produced by the experiments in this study are released as a single, self-contained data archive. The archive mirrors the directory layout used during training, so every artifact can be located deterministically from the tuple

(dataset, num_cells, quality, algorithm, signal, seed).

The full dataset is publicly mirrored on Amazon S3 with anonymous read-only access (no AWS credentials required):

```
s3://measurement-noise-scaling-laws/data/
```

The archive totals approximately 2.4 TB. A locally mounted copy is used for the experiments in this paper (resolved through `scaling_laws.paths.DATA_DIR`, default `~/noise_scaling/data`, overridable through the `NOISE_SCALING_DATA_DIR` environment variable).

2 Top-level layout

The archive contains four single-cell datasets, a shared protein-embedding bundle, and Weights & Biases experiment logs:

```
data/
|-- PBMC/           (~276-281 GB) CITE-seq peripheral blood mononuclear cells
|-- larry/          (~347-360 GB) State-fate lineage tracing
|-- merfish/        (~197 GB)   MERFISH spatial transcriptomics
|-- shendure/        (~1.6 TB)   Embryo developmental atlas (~10M cells)
|-- other/
|   '-- esm/         Shared ESM-2 / TranscriptFormer protein embeddings
|       |-- merged_esm_embeddings.pt
|       |-- required_gene_names.pkl
|       '-- coverage_report.txt
'-- wandb/           (~558 MB)   Weights & Biases experiment logs
```

3 Datasets

Four datasets span lineage, spatial, and developmental contexts (Table 1). Each dataset is sampled at 10 logarithmically-spaced training sizes and downsampled to 10 logarithmically-spaced quality levels (Table 2). Quality is defined as the fraction of original UMI counts retained after Numba-accelerated multinomial downsampling, so $\text{quality} = 1.0$ corresponds to no induced measurement noise.

Table 1: Datasets, biological signals, sources, and approximate scale.

Dataset	Description	Source	Total cells	Signal columns
PBMC	CITE-seq PBMCs	10x Genomics / Seurat	~100,000	celltype.13, protein_counts
larry	State-fate lineage tracing	Klein lab	~60,000	clone
merfish	MERFISH spatial transcriptomics	Allen Institute	~60,000	cur_idx, ng_idx
shendure	Embryo developmental atlas	Shendure lab / cellxgene	~10,000,000	author_day

Table 2: Training subset sizes (`num_cells`) and quality ranges per dataset. Each dataset uses 10 log-spaced sizes and 10 log-spaced quality levels.

Dataset	Sizes / quality range
PBMC	100, 215, 464, 1,000, 2,154, 4,641, 10,000, 21,544, 46,415, 100,000 quality $\in [0.0012346, 1.0]$
larry	100, 215, 464, 1,000, 2,154, 4,641, 10,000, 21,544, 46,415, 100,000 quality $\in [0.003876, 1.0]$
merfish	100, 203, 414, 843, 1,716, 3,494, 7,113, 14,480, 29,475, 60,000 quality $\in [0.027248, 1.0]$
shendure	100, 359, 1,291, 4,641, 16,681, 59,948, 215,443, 774,263, 2,782,559, 10,000,000 quality $\in [0.004, 1.0]$

4 Per-dataset directory structure

Every dataset directory follows an identical hierarchy. At the root sit the unprocessed AnnData object, dataset-level utility tables (gene-to-token mapping, gene medians, Ensembl mapping, HVG mask, t-digest normalization statistics), and held-out evaluation splits. Below this, each training subset of size `{num_cells}` contains 10 `{quality}` subdirectories, and each quality directory holds preprocessed AnnData, tokenized HuggingFace Arrow datasets, and per-algorithm trained-model directories with embeddings and MI estimates.

```
{dataset}/
|-- raw/raw.h5ad                # Original unprocessed AnnData
|-- utils/
|   |-- token_dict.pkl          # Geneformer gene -> token mapping
|   |-- gene_median_dict.pkl    # Per-gene expression medians
|   |-- detected_gene_median_dict.pkl
|   |-- ensembl_mapping_dict.pkl # Gene name -> Ensembl ID
|   |-- pca_hvg.pkl             # HVG boolean mask (750 genes)
|   |-- total_gene_tdigest_dict.pkl # T-digest normalization stats
|-- test/ validation/          # Same layout as training subsets
'-- {num_cells}/
    |-- sample.h5ad             # Subsampled cells (pre-downsampling)
    '-- {quality}/
        |-- sample.h5ad
        |-- preprocessed/
        |   |-- preprocessed.h5ad # Preprocessed + downsampled AnnData
        |   |-- lengths.pkl
        |   |-- tokenized.dataset/ # HuggingFace Arrow (Geneformer)
        '-- results/
            |-- Geneformer/{checkpoint-*, model/}
            |-- SCVI/model/
            |-- State/model/
            |-- PCA/model/
            '-- RandomProjection/model/
```

5 Embedding algorithms and trained artifacts

Five embedding algorithms are evaluated at every (*dataset*, *num_cells*, *quality*) combination. Each method’s trained model, test-set embeddings, and downstream MI estimates are stored in a parallel subtree under **results/** (Table 3).

Table 3: Embedding algorithms, their on-disk model file, embedding dimensionality, and short description.

Algorithm	Model file	Dim.	Description
Geneformer	<code>model.safetensors</code>	256	BERT-style masked language model over rank-encoded gene expression
SCVI	<code>model.pt</code>	16	Variational autoencoder with ZINB likelihood (scvi-tools)
STATE	<code>checkpoints/*.pt</code>	256	Self-supervised transformer encoder (Arc Institute); uses TranscriptFormer / ESM-2 protein embeddings from <code>other/esm/merged_esm_embeddings.pt</code>
PCA	<code>pca_model.pkl</code>	256	Truncated SVD on HVG-selected genes
RandomProjection	<code>random_projection.joblib</code>	256	Gaussian random projection baseline

STATE results are not produced for every (*size*, *quality*) combination: full coverage exists only for **merfish**, while **PBMC**, **larry**, and **shendure** contain STATE results only at their largest training size and quality = 1.0 (46,415, 46,415, and 59,948 cells respectively).

6 Shared ESM-2 / TranscriptFormer protein embeddings

`other/esm/merged_esm_embeddings.pt` is a shared 2,560-dimensional protein-language-model embedding table covering both *Mus musculus* and *Homo sapiens*, derived from TranscriptFormer’s `mus_musculus_gene.h5` and `homo_sapiens_gene.h5` releases and indexed by both Ensembl IDs and short gene names. STATE pre-processing auto-discovers this table and substitutes it for one-hot gene tokens. Per-dataset coverage against `var_names` of `raw/raw.h5ad` is reported in Table 4; `coverage_report.txt` in the same directory contains the full per-dataset breakdown, and `required_gene_names.pkl` stores the union of gene names across datasets.

Table 4: ESM-2 / TranscriptFormer protein-embedding coverage per dataset.

Dataset	Genes	In ESM dict	Coverage
PBMC	20,729	14,899	71.9%
larry	25,289	18,146	71.8%
merfish	483	479	99.2%
shendure	45,525	21,820	47.9%

7 Mutual-information estimates

Mutual information between embeddings and biological-signal columns is estimated with the latent mutual information (LMI) method using 4 random seeds: {42, 1404, 2303, 2701}. Each MI run produces three files (Table 5) stored at

`results/{Algorithm}/model/MI/{seed}/Y_{signal}_{quality}_{method}/`

Table 5: Files produced per MI run.

File	Description
<code>lmi_mutual_information.txt</code>	Scalar MI estimate (nats)
<code>lmi_embeddings.npy</code>	Optimized low-dimensional projection (NumPy array)
<code>lmi_model.pt</code>	Trained LMI neural-network state dict

Table 6: On-disk file formats and recommended loaders.

Extension	Format	How to load
<code>.h5ad</code>	AnnData (HDF5)	<code>anndata.read_h5ad(path)</code>
<code>.csv</code>	Comma-separated embeddings / signals	<code>pandas.read_csv(path, index_col=0)</code>
<code>.npy</code>	NumPy array	<code>numpy.load(path)</code>
<code>.pt</code>	PyTorch state dict / model	<code>torch.load(path)</code>
<code>.safetensors</code>	HuggingFace SafeTensors	<code>safetensors.torch.load_file(path)</code>
<code>.pkl</code>	Python pickle	<code>pickle.load(open(path, 'rb'))</code>
<code>.joblib</code>	Joblib-compressed object	<code>joblib.load(path)</code>
<code>.arrow</code>	Apache Arrow (HuggingFace datasets)	<code>datasets.load_from_disk(parent_dir)</code>

8 File formats

9 Programmatic and command-line access

The `S3Retriever` class (`scaling_laws.s3_retriever`) provides a unified interface to either the public S3 mirror or a local copy of the archive. It transparently caches S3 downloads to `~/.cache/noise_scaling_laws/` and exposes loaders for embeddings, MI scalars, LMI projections, AnnData samples, and trained models for every algorithm.

```
from scaling_laws.s3_retriever import S3Retriever

# Default: public S3 bucket
data = S3Retriever()

# Or point at a local copy
data = S3Retriever("/path/to/local/data/")

mi = data.load_mutual_information("PBMC", num_cells=1000, quality=1.0,
                                algorithm="Geneformer",
                                signal="celltype.l3", seed=42)
emb = data.load_embeddings("merfish", num_cells=7113, quality=1.0,
                           algorithm="Geneformer")
adt = data.load_sample("PBMC", num_cells=1000, quality=0.225772)
df = data.collect_all_mi_results("merfish")
```

The same module is exposed as a CLI (`python -m scaling_laws.s3_retriever {list,mi,collect,download} ...`). Direct `aws s3` access works without credentials by passing `--no-sign-request`:

```
aws s3 ls s3://measurement-noise-scaling-laws/data/ --no-sign-request

aws s3 sync s3://measurement-noise-scaling-laws/data/merfish/ ./merfish/ \
--no-sign-request

aws s3 cp s3://measurement-noise-scaling-laws/data/PBMC/1000/1.0/results/\
Geneformer/model/MI/42/Y_celltype.l3_1.0_geneformer/lmi_mutual_information.txt \
- --no-sign-request
```

10 Reproducibility

Because every artifact is keyed by the explicit tuple $(dataset, num_cells, quality, algorithm, signal, seed)$, any individual experiment in this paper can be located, downloaded, and re-evaluated independently of the rest of the archive. The full preprocessing-to-MI pipeline that generated this layout is described in the main repository `README` and orchestrated by the `Experiments` class in `scaling_laws.prepare.data`.